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TITLE PAGE

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Problem Statement ID – SIH26165

Problem Statement Title – AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act / Unsafe-Condition and Near-Miss Reports

Theme – Smart Automation

PS Category – Software

Team ID – ____

Team Name – Signal-0

SANKET

SIF Precursor Detection

Reads a free-text safety report and answers one question: could this have killed someone?

HAZARD → CONTROL STATUS → PLAUSIBLE SEVERITY

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SANKET — EARLY SAFETY WARNINGS FROM INCIDENT NARRATIVES

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Reads a safety or near-miss report and answers one question: could this have killed someone?

WHAT IT IDENTIFIES

- Hazard + Life-Saving Rule
- Control status
- Severity (1-5)
- SIF Precursor: Yes / No
- Evidence & reasoning

WHAT MAKES IT DIFFERENT

- Three gates, not one score
- One-Change severity rule
- LLM + local ML fallback
- Ranks a queue - never closes a report
- Every judgement logged

THE THREE GATES

GATE 1 · HAZARD

Is a high-energy hazard present?

yes · no · insufficient

→

GATE 2 · CONTROL

Was a barrier doing its job?

absent · failed · present · unclear

→

GATE 3 · SEVERITY

Plausible worst case?

1 - 5

SIF PRECURSOR = Hazard YES + Control ABSENT / FAILED + Severity >= 4

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TECHNICAL APPROACH

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CLIENT

Field Worker · React
voice or typed report → instant classification

HSE Admin · React
triage queue · dispatch / archive · risk board

↓

API

FastAPI
REST · Pydantic-validated

Classifier boundary
cache → primary → retry → fallback

Aggregation
plain SQL · GROUP BY

↓

MODELS

PRIMARY · Groq gpt-oss-20b
three-gate prompt · JSON out · hard deadline

FALLBACK · TF-IDF + LogReg
local · no network · answers when the API cannot

↓

DATA

reports
150 synthetic · 30 OSHA

predictions
append-only log

gold_labels
173 human labels

report_status
triage state

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FEASIBILITY AND VIABILITY

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FEASIBLE BECAUSE

- Uses incident reports that already exist
- No specialized hardware
- API-based & scalable
- LLM + local fallback
- Human-in-the-loop

CHALLENGE → SOLUTION

- Unstructured text → LLM analysis
- Ambiguous controls → 4-state classification
- API failure → retry + local fallback
- Domain variation → 30 real OSHA reports
- Model changes → one version per dashboard

OUR REAL LIMIT IS TOKENS, NOT COMPUTE

200,000
tokens / day, free tier

~3,150
tokens per report

63
reports / day

~2.8 days
full corpus re-run

SO THE PIPELINE IS BUILT TO SURVIVE IT

Resumable · never stores a fallback as if it were a real answer · dashboard always has data to show

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IMPACT AND BENEFITS

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POTENTIAL IMPACT

- Early detection of SIF precursors in OIL reports
- Prioritized review of high-risk incidents
- Ranks sites by precursor RATE, not count
- Finds recurring hazards across sites & shifts

BENEFITS

- Safety — act before the incident
- Operational — no manual first-pass screen
- Economic — prevents injuries and downtime
- Scalable — ten reports or ten thousand

Safety Reports → SIF Precursors → Risk Prioritization → Early Action → Safer Operations

WHY WE RANK BY RATE, NOT BY COUNT

A site that reports honestly logs more incidents. Counting them punishes it.

Count → the busiest site always looks worst

Rate → precursors ÷ reports, small groups held back

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RESEARCH AND REFERENCES

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RESEARCH

IOGP Life-Saving Rules · DEKRA SIF Framework · EEI SIF Model

DATA

150 synthetic reports + 30 real OSHA severe-injury narratives · 173 gold labels · 2 annotators

PROTOTYPE EVALUATION

HUMAN CEILING

0.947 Cohen's kappa

97.4% agreement · independent · n=38

TF-IDF BASELINE

0.754 F1 ± 0.077

5-fold CV · PR-AUC 0.847 · n=114

OSHA GENERALISATION

0.118 F1

real narratives · n=30, too small to conclude

LLM F1 withdrawn — we found a held-out report inside our own prompt and pulled the number rather than ship it. Re-measurement in progress.

STATED UP FRONT, BECAUSE A JUDGE WILL FIND THEM

Precursor rate 58.7%
not the 20–25% cited — our generator centred every report on a hazard

Ceiling is n=38
a re-label subset, so we quote the subset, never a bare kappa

No accuracy figure
and never a baseline scored on its own training data

GitHub: SANKET — AI-powered SIF precursor detection